

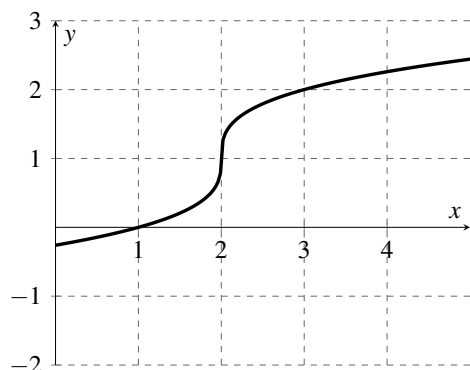
Name: _____

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Please circle your section: 901 (Gossell) 902 (Gossell) 903 (Graves-McCleary)

No aids (book, calculator, etc.) are permitted. **Show all work for full credit.**

1. [10 points] Consider the graph of
- $f(x) = \sqrt[3]{x-2} + 1$
- given below:



- a. Find the equation of the **secant line** passing through the points $(2, 1)$ and $(3, 2)$.
- b. Sketch the secant line from part (a) on the graph above. Label it “Secant Line”.
- c. The table below lists the slope (m_{sec}) of the secant line passing through the point $(3, 2)$ and the point $(x, f(x))$ for several values of x .
- | | | | | | | |
|-----------|---------|---------|---------|---------|---------|---------|
| x | 2.9 | 2.99 | 2.999 | 3.001 | 3.01 | 3.1 |
| $f(x)$ | 1.96549 | 1.99666 | 1.99967 | 2.00033 | 2.00333 | 2.03228 |
| m_{sec} | 0.34511 | 0.33445 | 0.33344 | 0.33322 | 0.33278 | 0.32280 |
- Use the information in the table to **estimate** the slope of the **tangent line** to $f(x)$ at $(3, 2)$.
- d. Use the slope from part (c) to write the equation of the tangent line at $(3, 2)$.
- e. Sketch the tangent line from part (d) on the graph above. Label it “Tangent Line”.

2. [7 points] The height of a hot air balloon after t minutes is given by $h(t) = 1500 - \frac{3000}{1+2^t}$ feet.

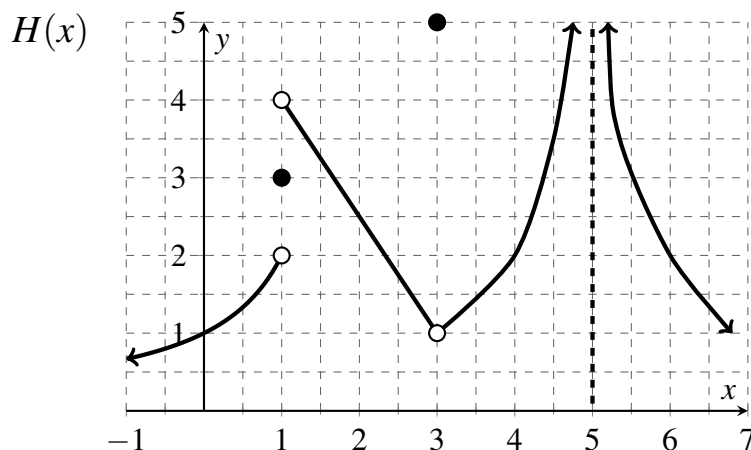
a. Find the height of the hot air balloon at time $t = 0$, $t = 1$, and $t = 2$.

$$h(0) = \underline{\hspace{2cm}} \qquad h(1) = \underline{\hspace{2cm}} \qquad h(2) = \underline{\hspace{2cm}}$$

b. Find the **average velocity** of the hot air balloon during the first minute (from $t = 0$ to $t = 1$).

c. Find the **average velocity** of the hot air balloon during the next minute (from $t = 1$ to $t = 2$).
Based on this, do you think the balloon is speeding up or slowing down?

3. [8 points] Use the graph of the function $H(x)$ (drawn below) to answer the questions. Assume $H(x)$ has a vertical asymptote at $x = 5$. For each problem below, give the most complete answer; if the limit is infinite, indicate that with $+\infty$ or $-\infty$. If the limit does not exist, write DNE.



a. $H(1) = \underline{\hspace{2cm}}$ b. $\lim_{x \rightarrow 1^-} H(x) = \underline{\hspace{2cm}}$ c. $\lim_{x \rightarrow 1^+} H(x) = \underline{\hspace{2cm}}$ d. $\lim_{x \rightarrow 1} H(x) = \underline{\hspace{2cm}}$

e. $\lim_{x \rightarrow 2} H(x) = \underline{\hspace{2cm}}$ f. $H(3) = \underline{\hspace{2cm}}$ g. $\lim_{x \rightarrow 3} H(x) = \underline{\hspace{2cm}}$ h. $\lim_{x \rightarrow 5} H(x) = \underline{\hspace{2cm}}$